

NexaCare Solutions: Non Functional Requirements Kit

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Project: Appointment Scheduling Platform – Version 3.0 Release

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Project scenario

NexaCare Solutions is a healthcare technology company supporting over 300 clinics across North America. The organization is preparing to launch Version 3.0 of its web-based appointment scheduling platform.

This version introduces new features that aim to improve usability, patient engagement, and clinic operations. As the Systems Professional, your responsibility is to coordinate and document a complete testing and release plan to ensure a safe, compliant, and successful rollout.

Here is the finalized **Section 8: Non-Functional Requirements and Performance Plan** for NexaCare Solutions, structured into the three specific tables required by your project documentation.

Section 8: Non-Functional Requirements and Performance Plan

NFR Definition

This table outlines the performance, scalability, and reliability requirements for the NexaCare platform version 3.0.

NFR Category	Specific Requirement	Target Value	Measurement Method
Performance	Appointment booking completion time	< 3 seconds	API response time monitoring
Scalability	Peak concurrent user capacity	2500 users	Load testing validation
Reliability	System uptime targets	99.9%	Service level

NFR Category	Specific Requirement	Target Value	Measurement Method
		Availability	monitoring logs
Availability	Maximum acceptable downtime during release	6 hours	Maintenance window tracking +1
Latency	SMS delivery timing	< 2 minutes	Transactional log timestamps

Golden Signals Monitoring Plan

The following plan identifies the Four Golden Signals for NexaCare and how they will be monitored to ensure system health.

Golden Signal	Metric Name	How to Measure	Target/ Threshold	Business Impact
Latency	API response time P95	Application performance monitoring	< 500 ms	Slow responses cause booking abandonment
Traffic	Active concurrent users	Real-time user sessions	Monitor for > 2500	Capacity planning and scaling triggers
Errors	HTTP 5xx error rate	Error tracking system	< 0.5%	Failed bookings = lost appointments
Saturation	DB connection pool utilization	Database monitoring	< 80%	Connection exhaustion causes failures

Peak Traffic Preparation

This table addresses requirements for the Monday morning peak traffic (3–5x normal volume) from 8:00 AM to 10:00 AM EST.

Metric	Normal Threshold	Peak Period Threshold	Reasoning
Database CPU	> 85% alert	> 95% alert	Higher utilization expected during peak hours
API Response Time	> 2s alert	> 4s alert	Accept slightly degraded performance vs. false alarms
Active Nodes	Minimum 3 servers	Minimum 6 servers	Horizontal scaling required to handle 2500 users
SMS Queue Lag	< 10 seconds	< 5 minutes	Accommodate high volume of 1h reminders for Monday clinics